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Advanced Calculus with Generalizations: Second Semester

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To the Instructor

0.1 Introduction - The Topic

In a way, Advanced Calculus is an odd topic. There seems to not be anything new going on, as by the time a student gets to this class he/she has had at least three semesters of calculus. So everyone is familiar with limits, differentiation, integration, sequences, and series. It is because the material is familiar that the study of analysis can be misleading. It is the depth that makes the course what it is and oftentimes confusion over the depth can create issues.

I still recall sitting in this class as a student wondering how in the world I was going to *prove* that absolute value was non-negative when there was nothing to do. Everyone just knows it is so. Such is how I began to realize I was not in Kansas anymore. However, I was determined to conquer this class and understand the deeper parts of calculus. In fact for me it was life changing as I went on to earn my PhD in classical real analysis and continue to publish in this area.

In this anecdote lie important details. After so many years being told that something was true, it was difficult to think along the lines of why it was true, especially since class at that time was all about listening to lecture, a passive act. However, there is something to get excited about as one sees that this is not a topic that has been wrung dry, but that there are generalizations and extensions to these topics which the interested student can follow to who knows where. There are ideas and proofs out there yet to be written. So this text was written with the goal of getting the students actively involved in thinking about mathematics and seeing some new(-ish) ideas out there.

0.2 Introduction - The Book

This book is for the second semester of a yearlong course in advanced calculus/real variables. It was used this way at Slippery Rock University where the sequence is a requirement for all math majors. Typically, such a class has between 8 and 20 students. The notes are parsed out to the students as a series of handouts. The pace of the course and the context of the handouts

will vary from instructor to instructor - you should edit the problems based on the ability and needs of your students. During the refereeing process I was asked to place back the pagebreaks when I developed these notes. This is a request I respectfully decline. You may decide to devote more time/problems to a topic and skip other things entirely. The L^AT_EX code is available for anyone who wants to shift things around.

As I teach the course, each handout contains Definitions, Axioms, Problems, Exercises, and a few Theorems. Anything labeled Theorem (which by virtue of being called a theorem must be true) needs to be proven. Any Problem will start with the directive “Prove or Disprove”. It is up to the student to determine the truthfulness of the statements and, if it is true, provide a proof; if it is false, give a counterexample with explanation. Both of these types of assignments can be presented at the board for points, and professionally written solutions are to be turned in. Exercises can be presented at the board for a grade, but are not turned in. The Exercises are to help illustrate definitions and topics. Similarly, there are Remarks throughout to explain concepts and introduce new discussions. Some Prove or Disprove items have quick and easy counter-examples. I have found that once the counter-examples are presented a good next step is to ask the student how to change the problem to make it into a theorem. Problem X then leads to Problem X, Part 2 which a student can then get credit for proving and presenting.

There is debate among practitioners of the Inquiry-Based Learning (IBL) about how to present the method to a class. Does one just plug away at the course, acting as if IBL is the usual thing or spend time explaining to the students how self-discovery is a better way to learn and own the material? Should the students be shown evidence of why IBL is best to help convince them that it is good? The decision of what to tell students is for most a personal matter. Each instructor must determine his/her own way. As for me, I steal an analogy from Ron Taylor of Berry College. At the beginning of the first day of class for the year, I talk about juggling. I define what juggling is and then give a short demonstration of juggling. Then I say to them, “Now you should be able to juggle. If you go home tonight and practice a little, you will be able to juggle just fine.” This is, in essence, the lecture-homework-move on format that most classes follow. Finally I pass out a review sheet of problems. These come from our introduction to proof course called Modern Concepts of Mathematics. There are two purposes to this: first, to review what I expect the students to know and be able to handle (induction, proof by contradiction, direct proof, etc.) and second, to get them up at the board on the second day, presenting something they should be more comfortable with as it is not new material. These problems, and these problems only, may be worked on in a group. The end of the second day is used to discuss the first page of Axioms, Definitions, Exercises, and

Conjectures. Of course, the first day of the second semester the students are already prepped on how the class runs. A group email before the first day can give them some problems to be thinking on for the new semester.

0.3 The Instructor in the Classroom

The most difficult part of being the instructor in an IBL setting is keeping quiet. This does not mean I remain absolutely quiet. Students should be encouraged to ask questions of the presenter and, especially at the start of the course, they sometimes need help formulating an exact question. So I will rephrase things or ask them to be more specific with their thoughts. In addition, if a presenter is having trouble, perhaps a fellow student has pointed out an issue that the presenter is struggling with resolving, I will suggest he/she take a break and we revisit the problem the next class day.

The instructor is also the official keeper of the records. I have the official list of which problems have been presented and which are still open. These are two things the students do tend to keep track of, too. Additionally, I keep track of any tweaks resulting in a new problem. The latter is something I really enjoy, though it took some getting used to. When new conjectures arise I do not always know the answer (as opposed to using a text where the problems are mostly the same regardless of the book). I learned to embrace the time of discovery via students. A typical day starts by asking if anyone has anything to present. Using my records, if I do see someone is falling behind in presentations I will say, “We haven’t heard from Z in awhile. Do you have something to present?” Some colleagues use software to keep track of such things. If you get involved in the IBL community you should be able to find them if you want.

At the beginning of the day I pass back write-ups, make a few general announcements (“Math Club meeting Thursday,” or “Make sure you are using the right notation in your limit proofs.”), and pass out any new pages. Then I take my seat *in the back of the room*. Students have a tendency to look at me to see if I react and use that to decide if they should or should not be looking for questions to ask. So the sides of the room are out. The presenter, the one person who is supposed to be looking in my direction, will still try and find confirmation in my face. It is important to remain impassive and let the presenter and his/her classmates work it all out.

After I sit down my role is that of moderator, helping to *further* the class discussion, but not to lead it. In the early days, I make sure the presenters are writing clearly and pausing between important statements so their classmates can keep up. I might spur students to ask questions by saying, “Does everyone agree with line 3?” or “Is there an assumption being made here when the theorem is applied?”

In summary, the instructor acts as bookkeeper with the problems and the grades and after that is a facilitator, making sure the class remains on task and that learning occurs.

0.4 Why these notes

One reason I decided to write my own notes is to introduce students to some of the generalizations and types of questions that someone like myself, who publishes in classical real analysis, faces in the scholarly life. Thus there are topics on these handouts that are not found in the typical analysis book/course. These include generalized continuity and convergence in metric spaces. None of these topics are covered too deeply (although metric spaces turn up again and again). It is amazing how well the students can do when they are not told that these topics are post-Bachelor's level or atypical.

0.5 Course Content

These notes are for the second semester of a two semester Advanced Calculus/Real Analysis sequence. There are several goals at work here. First is the in-depth study of a particular topic. Such depth is important for students of mathematics. Sometimes this is accomplished through taking an analysis sequence; sometimes a year-long algebra class; sometimes via two different, but related classes (a semester of algebra and one of number theory or a semester of real variables followed by one in complex variables). Second is honing the students' skills in writing proofs. All students by the time they are ready for this class have some learning in constructing proofs and proof techniques (direct, contradiction, contrapositive, induction, et al.), but one only gets better at this by writing more and more proofs, not by reading about proofs. Lastly, as with many "pure" math classes, this course should give students the opportunity to see how one extends the boundaries of what is known and not known via conjecture. This is where many textbook oriented classes fail as books do not rightly convey the stops and starts of what a mathematician studies, instead presenting a polished result that unfortunately gives students the impression that all that can be known is known.

The topics previously covered are

- fundamentals of the real line,
- limits and convergence, and
- functions and continuity.

In this second half, we continue onward by studying

- differentiation,
- integration,
- series of numbers, and
- series of functions.

Included in this text are several types of generalization of the main topics. For instance, symmetric derivatives and the Riemann-Stieltjes integral. The hope is to show students how things progress after what they learned in the Calculus sequence. Problems that are shown false by a counter-example (e.g. Let f have the property that $f(a) \cdot f(b) < 0$. Then there is a value c , with $a < c < b$ such that $f(c) = 0$.) are ripe for the question, "How can we change this to make it true?" during the problem's presentation. This shows how mathematicians hone an idea until it is both precise and correct. One of my favorite ideas is the so-called pathological example, something so far

afraid from a certain property as to be considered a “mathematical monster.” A continuous, nowhere differentiable function is an example of this. Many of the requests for examples in this text are leading the student toward some of these monsters.

0.6 Grading

My gradebook consists of three separate sections: homework, presentations, and exams. They are described below, however, for each instructor, there are many ways to determine grades. The ensuing paragraphs include the final percentages I use in this class. My technique is not a command, it’s not even a suggestion. It is an example.

Section One is for written homework turned in. This assigned homework is worth 20% of the final grade. Someone once referred to this method as having perpetual homework. No one warned me about the perpetual grading. You are now forewarned. I insist that homework is written up neatly and professionally. Students do not need to use $\text{\LaTeX}$ (though I would not complain), but since there is an emphasis on learning to communicate mathematics, the students need to learn that doing sloppy work is *not* acceptable. Homework falls into two categories: before presentation and after presentation. Before presentation homework that is correct and well-written receives a grade of 10. Originally, my syllabus said homework that has issues will receive a grade between 1 and 9, but I found over time that unless something spectacular happened I only used the grades 9 (small error), 7 (essentially correct, but not a perfect presentation), 5 (on the way, but a lot wrong), and 3 (trying). The grade of 1 was very rare as students did not have a hard deadline and would wait until they had some confidence in their work before turning it in. So rather than 1 - 10, my grades are now 1, 3, 5, 7, 9, and 10. After the problem was presented, a student could still turn in a write-up for either a grade of 5 (correct) or 3 (mostly correct). At the date of an exam, I cut off the students from turning in any more problems from the sections covered. There is a bit of self-defense here. I do not want thirty problems from the beginning of the semester turned in at the end because students finally got around to writing their solutions to them.

Section Two is for student presentations. Percentage-wise this counts as 10% of the grade total. The grading scale is 2 points through 5 points. Two is the minimum since students do not really present unless they have some content and confidence in their work, so they have earned something. I do not recall anyone actually settling for a two, though, as this means there are some major flaws in the mathematics. So you may think of the 2 as a placeholder, which will be replaced with a grade for a subsequent presentation. Three is for an average attempt. The student has given correct ideas, with

some flaws in terms of presentation and boardwork. A presentation at the level of 4 is correct and has a bit of polish to it. Something that can be followed by the students. A 5 is for an exceptional job. An absolutely correct solution with an excellent presentation where the concepts are broken down well and understood by the class earns this grade. On rare occasion, the presenter surprises me with a direction I never saw coming, I will give that a 5.5 out of 5 for creativity.

For a topic such as this, I tend to look at trends more than averages. I would not compare a student with 5 presentations worth 20 points against one with 8 presentations and 30 points. For the final grade I want to look at the *weighted* total. If the grade is borderline what I will look at is trends, to see if the presentation grades improved over time.

Students who are not actively presenting are still capable of earning points while a presentation is happening. This is Section Three. Those grades are for Questions, Insights, and Contributions (Q, I, C). Each of these types of grades is a single point, but students can earn multiple points per day. A “Q” is a good question. An example of this is, “Is your assumption the same as saying it’s a Lipschitz function?” Questions such as, “Can you say why your second line works?” and “Should that be f' and not f ?” are good questions, and students should be encouraged to ask them, but are not worth credit. The “I” is for some mathematical insight, a comment such as, “So a Riemann integral is a Riemann-Stieltjes integral where $g(x) = x$.” The “C” for contribution is hard to nail down exactly, but as Supreme Court Justice said, “I know it when I see it.” These points are part of the presentation grade. A good question or comment is as worthwhile as a nice presentation.

In addition to all of this, I give three exams and a comprehensive final. For each exam I will also give them a list of problems whose proof or counter-example I expect them to know. A majority of the exam, which is all proof and example, comes from that list with usually one problem thrown in that they have not seen to separate the A’s from the B’s in my class. These are two different parts of a student’s final grade. The exam average is 50% of the course grade and the final is 20%.

Two things I wish to point out before finishing this section:

First, I have been criticized in the past for the percentages I use. The critics’ point is mostly that students can pass this class without doing a lot of presentation and/or turning in their own version of solutions rather than what their peers show at the board. I, myself, don’t see much of a problem with this. If a student knows enough of the material, then he/she knows enough of the material. I do believe my tests do have one or two problems that require enough creativity to separate the A’s from the B’s from the rest.

Second, the question comes up how all these sections and percents come together to make a final grade. I was given some very prescient advice

before I started this class and it was absolutely spot on and something I wish to pass to you. The final totals tend to self-segregate. Each semester, when I computed the final spreadsheet there were obvious chunks where the A's, B's, C's, D's, and F's lay.

0.7 The Rules for Presentations

Presentations will be scored in the following way

- Accuracy of the problem you present, including the guidelines below.
- Defense of your work, including following the guidelines below.

Points: You will be awarded a point in the appropriate category every time you contribute in one of the following ways.

- P presentation points awarded
 - 5 points for a correct presentation
 - 2 - 4 points for a presentation with error(s)
- Q asking a good question of the presenter
- I contributing a demonstration of a mathematical insight
- C an oral contribution other than the two categories above

Remember: the presenter should be the person to answer a question.

Guidelines for Your Presentation

- Write the problem on the board.
- State what method/theorem/idea you will use.
- Clearly explain each step.
- Do not use the words stupid, trivial, obvious, etc.

Guidelines for Defending Your Work

- You must answer your classmates and professors questions in a respectful manner.
- Do not use the words stupid, trivial, obvious, etc.
- You must try to answer every question raised.

- It is okay to say, “I’m not sure that I understand your question.” It is not okay to say, “Your question doesn’t make any sense.”
- Talk to the class, not to the board.

Guidelines for Criticism of Classmates Work

- You are to ask questions about your classmates work. Do NOT suggest another technique. In some cases, there may be more than one way to solve a problem.
- Do not use the words stupid, trivial, obvious, etc.
- You must ask questions in a respectful manner.
- It is okay to say, “Can you explain how you got from line 3 to line 4?” It is not okay to say, “Line 4 is wrong,” or “Line 4 doesn’t make any sense.”

To the Student

0.8 Introduction - Advanced Calculus

Advanced Calculus (or Mathematical Analysis) is one of the most important classes one takes as a math student. Unlike a class in, say, abstract algebra, where the main concepts (rings, groups, fields) are new, in this class we will look at topics you should be familiar with from your earlier calculus sequence (limits, differentiation, integration, and sequences & series). That is the good news. Since you are familiar with all these topics, what this class then is about is an increase in rigor. This is not about determining the integral of f , but proving what type of f is an integrable function. We will also, as is expected in higher-level mathematics, look at some generalizations of these ideas. This serves as an introduction to what many mathematicians do when researching a topic.

Now, a word about prerequisites. Most courses of this type require a C or better in a course on proof techniques. Let us be blunt here: A grade of C or better does *not* mean that you have some vague recollection of a course that had a lot of proofs in it. It means that you learned the material and are capable of both creating your own proofs and following proofs of others. The types of proof techniques you should be familiar with include, but are not limited to, induction, direct proof, proof by contradiction, and proof by exhaustion.

This is an extremely important course as it develops the ability to apply logic, which is what makes the math major so desirable to employers (see www.maa.org/careers/ for more details). It takes hard work, but it is worth it.

Now, all that said, let me add that mathematics is one of the most creative disciplines out there. There is very little difference between the imagination, hard work, and focus needed to paint a still life and to determine a proof. Note that I did not say paint the *Mona Lisa* and prove *Fermat's Last Theorem*. Just as you don't have to paint masterpieces in order to enjoy painting, you do not have to solve the greatest open problems in mathematics to enjoy the thrill and pride associated with developing a proof. Learning to find proofs and write proofs is a skill that comes with practice; a lot of prac-

tice. If you start to feel discouraged and like you are flailing and lost, I suggest you think about the man who did solve Fermat's Last Theorem and his thoughts on doing math.

I can best describe my experience of doing mathematics in terms of a journey through a dark unexplored mansion. You enter the first room of the mansion and it's completely dark. You stumble around bumping into the furniture, but gradually you learn where each piece of furniture is. Finally, after six months or so, you find the light switch, you turn it on, and suddenly it's all illuminated. You can see exactly where you were. Then you move into the next room and spend another six months in the dark. So each of these breakthroughs, while sometimes they're momentary, sometimes over a period of a day or two, they are the culmination of, and couldn't exist without, the many months of stumbling around in the dark that proceed them.

Andrew Wiles, Princeton University

Get ready and get excited to take a journey deeper into analysis than you have taken before. And enjoy the trip!

0.9 R.L. Moore and his Method

That student is taught the best who is told the least.

R.L. Moore, 1882-1974

[Dr. Moore] told us early on that he had no use for the university guidelines stating that we should expect three hours of outside class work for each hour in the classroom. He said he wanted us to think about his class all day, every day, to go to bed thinking about it, to wake up in the night thinking about it, to get up the next morning thinking about it, to think about it walking to class, to think about it while we were eating. If we weren't prepared to do that, he didn't want us in his class. It was also quickly evident that he meant what he said....

John Green, PhD, University of Texas, 1968, under R. L. Moore

The core of any course using the Moore Method (a type of Inquiry-Based Learning) is the understanding that people learn best by doing the work, not by being told the results. Moore developed his method in 1911 while teaching at the University of Pennsylvania and then took it with him to the University of Texas where he worked from 1920 until his retirement in 1969. The mathematics building at UT is named after him.

To begin with, a true Moore Method course has no book. Instead, the “book” for the course is written by the students as the semester/year goes on. That is, at the end of this course the collection of definitions, axioms, and results you have will be enough for a text. Each and every student is expected to do his/her own work. The use of outside sources (including, but not limited to, books, the internet, tutors, friends, classmates, and professors who are not me) is strictly forbidden.

You will be given handouts which contain Definitions, Axioms, Problems, Exercises, and a few Theorems. You are to provide proofs for anything labeled Theorem (which by virtue of being called a theorem is true). Any problem will start with the directive “Prove or Disprove.” It is up to the student to determine the truthfulness of the statements and, if it is true, give a proof; or if it is false, give a counterexample with explanation. Exercises can be presented at the board for a grade, but not turned in. The Exercises are to help illustrate definitions and topics. Similarly, there are Remarks throughout to explain concepts and introduce new discussions.

In addition to submitting written work, students will be presenting at the board. There will be little, if any, lecture in this class. On class days, you will be called on to present solutions. You may choose to show your work on any problem not yet presented in class. If you are not the first person chosen on a given day, you will not have your choice of all assigned problems. This

means that you may not get to present your first choice problem, so you should be prepared with solutions to more than one problem.

During student presentations, the rest of the class is encouraged (expected) to ask questions, and to think critically about the solution presented by their classmates. The students in the class have the job of determining the validity of arguments presented, and the instructor will occasionally allow incorrect solutions to stand in class. These incorrectly presented problems may appear on tests. While “Should that two be a three?” is a question, you should also ask “Can you explain how you went from Step 2 to Step 3”. Don’t worry, if you have that question, so do others and the presenter should be able to answer that question and others. Every once in awhile there are questions someone doesn’t know how to answer. It happens to everyone, so the presenter should not feel bad or embarrassed. The response of “I’m not sure, I’ll have to get back to you,” is absolutely fine, but *the speaker is then responsible for finding out the answer and showing it to the class*. It will not take you long to figure out how to be the speaker and how to be an active audience member. Then it’s just a matter of enjoying the learning.

A quote from Paul Halmos¹:

“Don’t just read it; fight it! Ask your own questions, look for your own examples, discover your own proofs. Is the hypothesis necessary? Is the converse true? What happens in the classical special case? What about the degenerate cases? Where does the proof use the hypothesis?”

and another

“Mathematics is not a deductive science – that’s a cliché. When you try to prove a theorem, you don’t just list the hypotheses, and then start to reason. What you do is trial and error, experimentation, guesswork.”

and a final one

“It is the duty of all teachers, and of teachers of mathematics in particular, to expose their students to problems much more than to facts.”

¹If you do not know who Paul Halmos was, you may find it worthwhile to look him up.

Note: As with any written work, this is not a solitary effort, but a collaboration between the author and references, colleagues, & the past. These notes were inspired by the notes posted on the JIBL site from Scott Beaver of Western Oregon State University. They were also influenced by Jacqueline Jensen-Vallin and Ron Taylor and their efforts to help me convert to Inquiry-Based Learning. That said, all typos and other manners of errors are mine and mine alone. If the reader finds one (or many) mistake(s), please bring it to my attention, especially if the mistake causes a problem to be impossible. In addition, if you have questions on how to use this material, please do not hesitate to contact me at robert.vallin@sru.edu

Chapter 1

Differentiation

Definition 1. Suppose f is a function defined on the interval $I = [a, b]$. We say f is differentiable at $x_0 \in (a, b)$ if

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

exists and is finite. If such a value exists, it is called the derivative of f at x_0 and is denoted by $f'(x_0)$. If f is differentiable at each $x_0 \in (a, b)$, then we say f is differentiable on (a, b) .

Remark 2. 1. The expression

$$\frac{f(x) - f(x_0)}{x - x_0}$$

is called a difference quotient.

2. Another difference quotient and limit used in the definition of derivative is

$$\frac{f(x_0 + h) - f(x_0)}{h}.$$

In this case, the derivative could be defined by

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

3. We are making the assumption that any given x_0 is an interior point of an interval $[a, b] \subset \mathcal{D}(f)$. This is to avoid needing one-sided limits.
4. The reader may also be familiar with Leibniz ($\frac{d}{dx}$) notation. It is equivalent to writing with primes and we believe it is fair to assume your instructor and classmates can follow you, if you are more comfortable with that notation.

Exercise 3. Show that each of the following are differentiable at the given point.

1. $f(x) = x^5, x_0 = 2.$
2. $g(x) = \sqrt{x}, x_0 = 1$
3. $h(x) = \begin{cases} x^2 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}, x_0 = 0$

Prove or Disprove 4. The value of $f'(x_0)$ is unique.

Prove or Disprove 5. The function f is differentiable at x_0 if and only if f is continuous at x_0 .

Prove or Disprove 6. If f is differentiable at x_0 , then the function cf is differentiable at x_0 for any constant c .

Prove or Disprove 7. If f and g are both differentiable at x_0 , then their sum, $(f + g)$, is differentiable at x_0 .

Remark 8. A third way to define the derivative is via sequences. This can be helpful in proving conjectures. We say f is differentiable at x_0 if for every sequence $(x_n) \subset I \setminus \{x_0\}$ with $x_n \rightarrow x_0$ we have

$$\lim_{n \rightarrow \infty} \frac{f(x_n) - f(x_0)}{x_n - x_0}$$

exists, is finite, and has the same value.

Theorem 9. (Product and Quotient Rules) If f and g are both differentiable at x_0 , then so are fg and f/g with

$$(fg)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0)$$

and

$$(f/g)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{[g(x_0)]^2}, \text{ assuming } g(x_0) \neq 0$$

Theorem 10. (Chain Rule) Let f and g be functions with the domain of f containing the range of g . If g is differentiable at x_0 and f is differentiable at $g(x_0)$, then $f \circ g$ is differentiable at x_0 and

$$(f \circ g)'(x_0) = f'(g(x_0)) \cdot g'(x_0).$$

Definition 11. Suppose f is a function on an open interval (a, b) . Then f has a local maximum (local minimum) at $x_0 \in (a, b)$ if there exists $\delta > 0$ such that for all $y \in (a, b) \cap (x_0 - \delta, x_0 + \delta)$, $f(y) \leq f(x_0)$ ($f(y) \geq f(x_0)$).

Theorem 12. Suppose f is a function on (a, b) . If f has a local maximum or a local minimum at $x_0 \in (a, b)$ and if f is differentiable at x_0 , then $f'(x_0) = 0$.

Theorem 13. (Rolle's Theorem) If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then there exists an $x_0 \in (a, b)$ such that $f'(x_0) = 0$.

Exercise 14. Show that if the hypothesis “differentiable on (a, b) ” is dropped from Rolle’s Theorem, the conclusion is false.

Theorem 15. (Mean Value Theorem) If f is continuous on $[a, b]$ and differentiable on (a, b) , then there exists an $x_0 \in (a, b)$ such that

$$f'(x_0) = \frac{f(b) - f(a)}{b - a}.$$

Exercise 16. Give an example to show that if f has even one point of discontinuity in (a, b) , then the conclusion of the Mean Value Theorem may not be true.

Exercise 17. Give an example to show that if f has even one point of non-differentiability in (a, b) , then the conclusion of the Mean Value Theorem may not be true.

Prove or Disprove 18. If f is differentiable on (a, b) with $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on (a, b) .

Definition 19. A function f is increasing (decreasing) on the interval (a, b) if for any $x, y \in (a, b)$ with $x < y$ we have $f(x) \leq f(y)$ ($f(x) \geq f(y)$). If the $\leq$ in the conclusion is replaced with $<$ we say f is strictly increasing (strictly decreasing).

Prove or Disprove 20. If $f'(x) \geq 0$ for all $x \in (a, b)$, then f is increasing on (a, b) .

Definition 21. Let f be a function. We say f is bounded on the interval (a, b) (a and b can be finite or infinite) if there exists real numbers m, M such that $m \leq f(x) \leq M$ for all $x \in (a, b)$.

Definition 22. Let $D \subseteq \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$. We say f is uniformly continuous if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that for every $x, y \in D$ with $|x - y| < \delta$ we have

$$|f(x) - f(y)| < \varepsilon.$$

Theorem 23. Let f be differentiable on the interval (a, b) with f' bounded. Then f is uniformly continuous on (a, b) .

Exercise 24. Show that $f(x) = \sin(x)$ is a uniformly continuous function on $\mathbb{R}$.

Exercise 25. Construct a differentiable, uniformly continuous function f such that f' is unbounded. This shows that it is not necessary that f' be bounded in order that f be uniformly continuous.

Remark 26. Recall the following two definitions from the first semester.

Let $X \subseteq \mathbb{R}$ and let $\{f_n\}$ and f be real functions with domain X . We say f_n converges pointwise to f if for each $x_0 \in X$ we have

$$\lim_{n \rightarrow \infty} f_n(x_0) = f(x_0);$$

that is, for every $x_0 \in X$ and every $\varepsilon > 0$ there exists an $N \in \mathbb{N}$ such that for $n > N$

$$|f_n(x_0) - f(x_0)| < \varepsilon.$$

We write this as $f_n \xrightarrow{p} f$. Additionally, we say f_n converges uniformly to f if for every $\varepsilon > 0$ there exists an $N \in \mathbb{N}$ such that if $n > N$ we have

$$|f_n(x) - f(x)| < \varepsilon$$

for all $x \in X$. We denote this by

$$f_n \xrightarrow{u} f.$$

Prove or Disprove 27. Let $f_n : (a, b) \rightarrow \mathbb{R}$ be a sequence of functions. If each f_n is differentiable on (a, b) and $f_n \xrightarrow{p} f$, then f is differentiable on (a, b) .

Definition 28. A function f is said to be continuously differentiable on an interval I if it is differentiable on I and the function f' is continuous on I .

Exercise 29. Find examples of functions f and g where f is continuously differentiable and g is differentiable, but not continuously so.

Theorem 30. (This is an exception to the rule. You need not prove this. It is included for completeness.) Let $\{f_n\}$ be a sequence of continuously differentiable functions defined on an interval I and let f and g be functions defined on I where $f_n \xrightarrow{p} f$ on I . If $f'_n \xrightarrow{u} g$ on I , then f is differentiable on I and $f' = g$.

Definition 31. A function f is said to be in Baire Class One if it is the pointwise limit of continuous functions, f_n , but is not itself continuous; that is, a discontinuous f is Baire Class One if there exists f_n , each of which is continuous, so that

$$f_n \xrightarrow{p} f.$$

Exercise 32. Find an example of a function which has a point of discontinuity, but is in Baire Class One.

Prove or Disprove 33. A function can have infinitely many discontinuities but be in Baire Class One.

Theorem 34. If f is a derivative (that is, there exists a differentiable function g such that $g'(x) = f(x)$ for all x), then f is in Baire Class One.

Theorem 35. Let f be differentiable on the interval I . Suppose $a, b \in I$ with $a < b$ and $f(a) \neq f(b)$. Let z be any number between $f'(a)$ and $f'(b)$. Then there exists a number c between a and b so that $f'(c) = z$.

Remark 36. This shows that every derivative has the Intermediate Value (Darboux) Property¹.

Prove or Disprove 37. There exists a function f on $\mathbb{R}$ such that

$$f'(x) = \chi_{\{0\}} = \begin{cases} 0 & x \neq 0 \\ 1 & x = 0 \end{cases},$$

the characteristic function of zero.

Definition 38. Let $f : \mathbb{R} \rightarrow \mathbb{R}$. We say that c is a sequential derivate of f at x_0 if there exists sequence $\{h_n\}$ converging to zero so that

$$\lim_{n \rightarrow \infty} \frac{f(x_0 + h_n) - f(x_0)}{h_n} = c.$$

Prove or Disprove 39. If f is differentiable at x_0 , then f has a unique sequential derivate at x_0 .

Exercise 40. Find an example of a function which is not differentiable at a point x_0 , but has sequential derivate(s) at x_0 .

Exercise 41. Find an example of a function f that has more than one finite sequential derivate at x_0 and is continuous at x_0 ?

Prove or Disprove 42. A function can have infinitely many sequential derivate values.

Theorem 43. Suppose that the functions f and g are continuous on some interval $[a, b]$ containing the point c and differentiable on (a, b) (except possibly at c itself). Furthermore, suppose that

1. $\lim_{x \rightarrow c} f(x) = 0$,
2. $\lim_{x \rightarrow c} g(x) = 0$,
3. for every $x \in (a, b) \setminus \{c\}$, $g'(x) \neq 0$, and
4. $\lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$ exists.

Then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}.$$

Remark 44. This is the first version of L'Hopital's Rule that you learn in Calculus I. There are other indeterminate forms (what are they?), but this text will not go into them.

¹Let $f : [a, b] \rightarrow \mathbb{R}$. We say f has the IVP if for any v , is a number between $f(a)$ and $f(b)$, there is a point $c \in (a, b)$ such that $f(c) = v$.

Chapter 2

Integration

Definition 45. A partition P of $[a, b]$ is a finite set of $n + 1$ elements of $[a, b]$ containing at least one element of (a, b) with

$$P = \{t_0 = a, t_1, t_2, \dots, t_{n-1}, t_n = b\}$$

where $t_{k-1} < t_k$ for $k = 1, 2, \dots, n$. If k is a natural number no greater than n , we will denote the k^{th} subinterval $[t_{k-1}, t_k]$ by I_k and let $\Delta I_k = t_k - t_{k-1}$. Define the mesh of P , written as $\text{mesh}(P)$, as

$$\text{mesh}(P) = \max_k \{\Delta I_k\}.$$

Definition 46. Let f be a function defined on $[a, b]$, and let P be a partition of $[a, b]$. Let

$$m_k(f) = \inf_{x \in I_k} \{f(x)\}$$

$$M_k(f) = \sup_{x \in I_k} \{f(x)\}.$$

Then we define the Lower and Upper Type I sums of f on $[a, b]$ relative to P by

$$L(f, P) = \sum_{k=1}^n m_k(f) \Delta(I_k)$$

$$U(f, P) = \sum_{k=1}^n M_k(f) \Delta(I_k).$$

Prove or Disprove 47. Let f be defined on $[a, b]$ and let P be a partition of $[a, b]$. Then $L(f, P) \leq U(f, P)$.

Definition 48. A partition Q is a refinement of the partition P if $P \subseteq Q$.

Exercise 49. Given two partitions P_0 and P_1 , find a partition that is a refinement of both.

Prove or Disprove 50. Let f be defined on $[a, b]$ and let P, Q be partitions of $[a, b]$ with Q a refinement of P . Then

$$L(f, P) \leq L(f, Q) \leq U(f, Q) \leq U(f, P).$$

Exercise 51. Let f be the characteristic function of the rationals on $[0, 1]$; that is

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{otherwise} \end{cases}.$$

Determine $U(f, P)$ and $L(f, P)$ where P is a partition of $[0, 1]$.

Exercise 52. Let $g : [0, 2] \rightarrow \mathbb{R}$ be the function

$$g(x) = \begin{cases} 2 & \text{if } x > 1 \\ 0 & \text{if } x \leq 1 \end{cases}.$$

Let P be a partition of $[0, 2]$ with $x_k < 1 < x_{k+1}$ for some index k . Determine $U(f, P)$ and $L(f, P)$.

Definition 53. Let f be defined on $[a, b]$. Then we define the Upper and Lower Type I integrals of f by

$$U(f) = \inf\{U(f, P) : P \text{ is a partition of } [a, b]\}$$

$$L(f) = \sup\{L(f, P) : P \text{ is a partition of } [a, b]\}.$$

If $U(f) = L(f)$, then we say f is Type I integrable on the interval and denote this number as $\int_a^b f$.

Exercise 54. Show that characteristic function of the rationals on $[0, 1]$ is not Type I integrable on $[0, 1]$.

Exercise 55. Show that $f(x) = x$ is Type I integrable on $[0, b]$ where $b > 0$.

Prove or Disprove 56. If f is a bounded function on I , then

$$L(f) \leq U(f).$$

Theorem 57. The function f is Type I integrable on $[a, b]$ if and only if for every $\varepsilon > 0$ there exists a partition P of $[a, b]$ such that $U(f, P) - L(f, P) < \varepsilon$.

Definition 58. A Type II sum associated with f on $[a, b]$ with respect to P is a sum of the form

$$T(f, n, P) = \sum_{k=1}^n f(x_k) \Delta(I_k)$$

where $x_k \in I_k$ for $k = 1, 2, \dots, n$. We say that f is Type II integrable on $[a, b]$ if and only if there exists a real number $T(f)$ such that for all $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$|T(f, n, P) - T(f)| < \varepsilon$$

for every Type II sum associated with f on $[a, b]$ with respect to P , where $\text{mesh}(P) < \delta$. If f is Type II integrable on $[a, b]$, we denote $T(f)$ as the Type II integral of f .

Theorem 59. The function f is Type II integrable on $[a, b]$ if and only if it is Type I integrable on $[a, b]$.

Remark 60. Since Type I and Type II integrals are equivalent and we will denote both by $\int_a^b f$.

Prove or Disprove 61. If f is integrable on $[a, b]$, then f is integrable on $[c, d]$ where $a < c < d < b$.

Prove or Disprove 62. If f is monotone on $[a, b]$, then f is integrable on $[a, b]$.

Prove or Disprove 63. If f is continuous on $[a, b]$, then f is integrable on $[a, b]$.

Exercise 64. Show the differences in Prove or Disprove 62 and 63. Find a monotone function on $[a, b]$ that is not continuous and a continuous function on $[a, b]$ that is not monotone.

Prove or Disprove 65. If f and g are integrable on $[a, b]$, then their sum $f + g$ is also integrable on $[a, b]$.

Prove or Disprove 66. If f is integrable on $[a, b]$ and c is any real number, then cf is integrable on $[a, b]$.

Prove or Disprove 67. If $f > 0$ on $[a, b]$ and integrable, then $\int_a^b f > 0$.

Prove or Disprove 68. If $f > g$ on $[a, b]$ (i.e. $f(x) > g(x)$ for all $x \in [a, b]$) and f, g are both integrable, then $\int_a^b f > \int_a^b g$.

Prove or Disprove 69. If f is integrable on $[a, b]$, then $|f|$ is integrable on $[a, b]$ and $|\int_a^b f| \leq \int_a^b |f|$.

Prove or Disprove 70. If $|f|$ is integrable on $[a, b]$, then f is integrable on $[a, b]$.

Theorem 71. If f is bounded and integrable on $[a, b]$, then f^2 is integrable on $[a, b]$.

Prove or Disprove 72. If f and g are integrable on $[a, b]$, then their product fg is integrable on $[a, b]$.

Prove or Disprove 73. If f and g are integrable on $[a, b]$, then $\max\{f, g\}$ and $\min\{f, g\}$ are integrable on $[a, b]$.

Prove or Disprove 74. If f is defined on $[a, b]$ and $c \in (a, b)$ and f is integrable on both $[a, c]$ and $[c, b]$, then f is integrable on $[a, b]$ and

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

Prove or Disprove 75. If f has finitely many discontinuities on $[a, b]$, then f is integrable on $[a, b]$.

Theorem 76. Let $f_n \xrightarrow{u} f$ on $[a, b]$ and suppose each f_n is integrable. Then f is integrable and

$$\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n.$$

Exercise 77. Give an example to show that uniform convergence cannot be replaced by pointwise convergence in the theorem above.

Theorem 78. (The Mean Value Theorem for Integrals) If f is continuous on $[a, b]$ then there exists a $c \in (a, b)$ such that

$$f(c) = \frac{1}{b-a} \int_a^b f.$$

(Also, draw a picture to give this a geometric meaning.)

Theorem 79. (The Fundamental Theorem of Calculus I) Let f be continuous on I and differentiable on (a, b) . Then if f' is integrable on $[a, b]$ we have

$$\int_a^b f' = f(b) - f(a).$$

Theorem 80. (The Fundamental Theorem of Calculus II) Let f be integrable on $[a, b]$. Then if $x \in [a, b]$, define

$$F(x) = \int_a^x f.$$

Then F is continuous on $[a, b]$. If f is continuous at $x_0 \in (a, b)$, then F is differentiable at x_0 and $F'(x_0) = f(x_0)$.

Definition 81. The sequence of functions f_n on $[a, b]$ converges discretely to f if for every $x \in [a, b]$ there exists an $N \in \mathbb{N}$ such that for $n > N$ we have $f_n(x) = f(x)$. We write this as $f_n \xrightarrow{d} f$.

Prove or Disprove 82. If f_n is integrable on $[a, b]$ and $f_n \xrightarrow{d} f$, then f is integrable on $[a, b]$.

Definition 83. Let P be a partition of $[a, b]$ and f a bounded function defined on $[a, b]$. Let g be an increasing function from $[a, b]$ into $\mathbb{R}$. Then define $\Delta g(I_k)$ by

$$\Delta g(I_k) = g(t_k) - g(t_{k-1})$$

for $i = 1, 2, 3, \dots, n$. We define the Lower and Upper Type I sums of f on $[a, b]$ relative to P with respect to g by

$$L(f, g, P) = \sum_{k=1}^n m_k(f) \Delta g(I_k)$$

and

$$U(f, g, P) = \sum_{k=1}^n M_k(f) \Delta g(I_k).$$

Exercise 84. Suppose P is a partition of the unit interval, f is the characteristic function of the rationals, and $g(x) = x^2$. Find the values of $L(f, g, P)$ and $U(f, g, P)$.

Definition 85. Let P be a partition of $[a, b]$ as earlier. Let g be a strictly increasing function from $[a, b]$ into $\mathbb{R}$. The Upper and Lower Riemann-Stieltjes integrals of f with respect to g over $[a, b]$ are defined as

$$\int_a^b f dg = \inf \{ U(f, g, P) : P \text{ is a partition of } [a, b] \}$$

and

$$\int_a^b f dg = \sup \{ L(f, g, P) : P \text{ is a partition of } [a, b] \}$$

Remark 86. When $\int_a^b f dg = \int_a^b f dg$, we say f is Riemann-Stieltjes integrable with respect to g and denote this number as $\int_a^b f dg$. We refer to the set of Riemann-Stieltjes integrable functions with respect to g by $\mathcal{R}_g$ or, when the interval is not understood, $\mathcal{R}_g[a, b]$.

Prove or Disprove 87. Let f be in $\mathcal{R}_g$ and let g_1, g_2 be increasing functions on $[a, b]$. If $g_1 \leq g_2$, then $\int_a^b f dg_1 \leq \int_a^b f dg_2$.

Prove or Disprove 88. Let f be in $\mathcal{R}_g$ and g an increasing function on $[a, b]$. Then for any constant c , $\int_a^b f d(cg) = c \int_a^b f dg$.

Exercise 89. Let $f(x) = c$ on $[a, b]$. Determine the value of $\int_a^b f dg$.

Exercise 90. Let $f \in \mathcal{R}_g[0, 3]$ where g is defined by

$$g(x) = \begin{cases} 2 & \text{if } x > 1 \\ 0 & \text{if } x \leq 1 \end{cases}$$

Determine the value of $\int_a^b f dg$.

Chapter 3

Series and Series of Functions

A Bit of Review.

Definition 91. A sequence is a function f with domain $\mathbb{N}$ and range a subset of $\mathbb{R}$.

Definition 92. The sequence (x_n) converges to the real number L , written

$$\lim_{n \rightarrow \infty} x_n = L,$$

or $x_n \rightarrow L$, if for every $\varepsilon > 0$ there exists a natural number N such that for all $n > N$ we have

$$|x_n - L| < \varepsilon.$$

Definition 93. A sequence which does not converge is said to diverge.

This can happen in several ways. Take a moment to consider how this can happen.

Exercise 94. Determine the negation of the definition of convergence for a formal definition we use when the limit does not exist.

Remark 95. There is also the possibility that the $\lim_{n \rightarrow \infty} x_n$ exists, but it is not finite.

Definition 96. We say $\lim_{n \rightarrow \infty} x_n = \infty$ if for every $M \in \mathbb{R}$ there exists a natural number N such that if $n > N$, then $x_n > M$.

Exercise 97. Formulate the analogous definition for $\lim_{n \rightarrow \infty} x_n = -\infty$.

Definition 98. We say a sequence $\mathbf{x}$ is bounded if there exists numbers m and M such that for all $n \in \mathbb{N}$

$$m \leq x_n \leq M.$$

Definition 99. Let $\mathbf{x}$ be a bounded sequence. Define sequences (u_n) and (v_n) by, for all $N \in \mathbb{N}$

$$u_N = \inf \{x_n : n > N\}$$

and

$$v_N = \sup\{x_n : n > N\}.$$

Example 100. Let $\mathbf{x} = \{1 + 1/2, 1/2, 1 + 1/3, 1/3, 1 + 1/4, 1/4, \dots\}$. Then $u_N = 0$ for all N while $v_0 = 1 + 1/2$, $v_1 = 1 + 1/3$, $v_2 = 1 + 1/3$, and $v_3 = 1 + 1/4$.

Definition 101. Let $\mathbf{x}$ be a bounded sequence and let (u_N) and (v_N) be defined as above. Define the limit inferior and limit superior of $\mathbf{x}$, denoted by $\liminf x_n$ and $\limsup x_n$, respectively, as

$$\liminf x_n = \lim_{N \rightarrow \infty} u_N$$

and

$$\limsup x_n = \lim_{N \rightarrow \infty} v_N.$$

Example 102. So for $\mathbf{x} = \{1 + 1/2, 1/2, 1 + 1/3, 1/3, 1 + 1/4, 1/4, \dots\}$, $\liminf x_n = 0$ while $\limsup x_n = 1$.

Definition 103. A sequence (x_n) of real numbers is called a Cauchy sequence if for every $\varepsilon > 0$ there exists a natural number N so that if $n > m \geq N$, then

$$|x_n - x_m| < \varepsilon.$$

Definition 104. A subsequence is a (infinite) sequence whose terms consist of some of the terms of (x_n) taken in order. The usual notation is to write our subsequence as

$$(x_{n_k}) = x_{n_1}, x_{n_2}, x_{n_3}, \dots$$

where n_1 denotes the first term taken from (x_n) , n_2 denotes the second term taken from (x_n) , et cetera. Notice it must be true that for the value of n_k we must have $n_k \geq k$.

On to New Stuff.

Definition 105. The symbol $\sum_{n=m}^{\infty} a_n$ means $a_m + a_{m+1} + a_{m+2} + \dots$. To give this some meaning, we change this into a sequence and consider

$$s_k = a_m + a_{m+1} + \dots + a_k = \sum_{n=m}^k a_n$$

for each $k \geq m$. The elements of $\{s_k\}$ form the sequence of partial sums. The infinite series $\sum a_n$ converges if and only if the sequence of partial sums $\{s_k\}$ converges to the real number S . Then we say

$$\sum_{n=m}^{\infty} a_n = \sum a_n = S.$$

If a series does not converge, then we say it diverges.

Theorem 106. Geometric Series

If $|r| < 1$ and a is any real number, then the series

$$\sum_{n=0}^{\infty} ar^n = a + ar + ar^2 + ar^3 + \cdots$$

converges.

Definition 107. p-Series

The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ where p is fixed.

Theorem 108. This series converges if and only if $p > 1$.

This type of series is notorious for being easy to determine convergence/divergence, but difficult for finding the exact sum. For example (without proof), the so-called Basel Problem was to find the exact value of

$$\sum_{n=1}^{\infty} \frac{1}{n^2}.$$

It was solved by Leonhard Euler in 1735 who showed

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Prove or Disprove 109. The series $\sum a_n$ converges if and only if $a_n \rightarrow 0$.

To determine convergence (but not the sum's value) we have many different tests.

Theorem 110. Comparison Test

Let $\sum a_n$ be a series where $a_n \geq 0$ for all n .

1. If $\sum a_n$ converges and $|b_n| \leq a_n$, for all n , then $\sum b_n$ converges.
2. If $\sum a_n = \infty$ and $b_n \geq a_n$, for all n , then $\sum b_n = \infty$.

Prove or Disprove 111. If $\sum |a_n|$ converges, then $\sum a_n$ converges.

Definition 112. A series is said to converge absolutely if $\sum |a_n|$ is a convergent series.

Prove or Disprove 113. Let $\{a_n\}$ be a sequence of nonzero real numbers such that a_{n+1}/a_n is a constant. Then $\sum a_n$ is a geometric series.

Theorem 114. Ratio Test

A series $\sum a_n$ of nonzero terms

1. converges absolutely if $\limsup |a_{n+1}/a_n| < 1$,
2. diverges if $\liminf |a_{n+1}/a_n| > 1$.

3. Otherwise, if $\liminf |a_{n+1}/a_n| \leq 1 \leq \limsup |a_{n+1}/a_n|$ the test fails and the series may converge or diverge.

Exercise 115. Find examples of two series, $\sum x_n$ and $\sum y_n$ such that $\sum x_n$ is convergent, $\sum y_n$ is divergent, and The Ratio Test fails for both of them.

Theorem 116. Root Test

Let $\sum a_n$ be a series and let $\alpha = \limsup |a_n|^{1/n}$. The series

1. converges absolutely if $\alpha < 1$,
2. diverges if $\alpha > 1$.
3. Otherwise $\alpha = 1$ and the test fails.

Exercise 117. Let a_n be a sequence defined by

$$a_n = \begin{cases} 2r^n & n \text{ odd} \\ r^n & n \text{ even} \end{cases}$$

where r is a fixed, nonzero number. Show that this is an example of when the Ratio Test fails, but the Root Test does not.

Theorem 118. Let $\sum_{n=1}^{\infty} a_n$ be a series with non-negative terms and let $f : [0, \infty) \rightarrow \mathbb{R}$ be a continuous function such that $f(n) = a_n$ for all n . Then the series converges if and only if

$$\lim_{t \rightarrow \infty} \int_1^t f(x) dx$$

exists and is finite.

Remark 119. Theorem 118 is the easy way to prove convergence for p -series.

Definition 120. Let $a_n > 0$ for all n . An alternating series is a series of the form

$$\sum_{n=1}^{\infty} (-1)^n a_n \text{ or } \sum_{n=1}^{\infty} (-1)^{n+1} a_n.$$

Theorem 121. An alternating series converges if and only if $a_n \rightarrow 0$.

Definition 122. Let $f : \mathbb{R} \rightarrow \mathbb{R}$. We say that f is convergence preserving if for every convergent series $\sum a_n$, the series

$$\sum f(a_n)$$

also converges.

Prove or Disprove 123. If f is convergence preserving, then $\lim_{x \rightarrow 0} f(x) = 0$.

Prove or Disprove 124. If f is convergence preserving, then f is continuous at $x_0 = 0$.

Prove or Disprove 125. If f is convergence preserving, then f is differentiable at $x_0 = 0$.

Definition 126. Given a sequence $\{a_n\}$ of real numbers, the series $\sum_{n=0}^{\infty} a_n x^n$ is called a power series. For any power series, one of the following is true:

1. The power series converges for all $x \in \mathbb{R}$;
2. The power series converges for only $x = 0$;
3. The power series converges for all x in some bounded interval centered at 0.

If instead of x^n we have $(x - c)^n$ this is referred to as a power series centered at c .

Theorem 127. Let $\sum a_n(x - c)^n$ be a power series. Define s as

$$s = \limsup_{n \rightarrow \infty} |a_{n+1}/a_n|$$

and let

$$r = \begin{cases} 0 & \text{if } s = \infty \\ 1/s & \text{if } 0 < s < \infty \\ \infty & \text{if } s = 0 \end{cases}$$

Then $\sum a_n(x - c)^n$ converges absolutely for all values of x that satisfy $|x - c| < r$ and diverges for all values of x that satisfy $|x - c| > r$. The number r is called the radius of convergence. The values of x for which the series converge are called the interval of convergence. From the theorem the interval contains $(c - r, c + r)$. The endpoints of the interval must be checked individually.

Exercise 128. Find the interval and radius of convergence for

$$\sum_{n=1}^{\infty} n^n (x + 4)^n.$$

Exercise 129. Find the interval and radius of convergence for

$$\sum_{n=0}^{\infty} \frac{10^n}{n!} (x - 3)^n.$$

Exercise 130. Find the interval and radius of convergence for

$$\sum_{n=1}^{\infty} \frac{(x - 1)^n}{n 3^n}.$$

Definition 131. Suppose $\sum_{k=0}^{\infty} a_k(x-a)^k$ has radius of convergence $r > 0$. Let $f(x) = \sum_{k=0}^{\infty} a_k(x-a)^k$ for each $x \in (x-r, x+r)$. Then we say f is represented by the power series $\sum_{k=0}^{\infty} a_k(x-a)^k$ or that $\sum_{k=0}^{\infty} a_k(x-a)^k$ is the power series representation for f .

Remark 132. We now look at going in the other direction; that is, if we start with a function f , can we always find a series representation for f , and if not, what conditions on the function are necessary for the series to exist?

Definition 133. Let f have derivatives of all orders on an interval I and let c be an interior point of I . The Taylor Series for f centered at c is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(c)}{k!} (x-c)^k.$$

When $c = 0$ this is called the Maclaurin Series for f .

Exercise 134. Find the Maclaurin Series for each of the following and find the interval of convergence:

1. $\frac{1}{1-x}$
2. e^x
3. $\sin(x), \cos(x)$

Remark 135. For a Taylor (Maclaurin) Series, the convergence of the partial sums is uniform. This means the properties of continuity, differentiability, and integrability are passed through limits; i.e. since each term is continuous, differentiable, and integrable (on closed intervals $[a, b]$), then so is the series.

Exercise 136. Find the Maclaurin series for each of the following by using Exercise 134 and Remark 135. Find the interval of convergence:

1. $\frac{1}{1+x}$
2. $\sinh(x), \cosh(x)$
3. $\ln(1+x)$
4. $\arctan(x)$